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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 03/23/2006 |
| **PI Name:** | Connolly, Janis M.S. | | |
| **Project Title:** | Concepts for Simplified Stowage | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES | | |
| **Program/Discipline--Element/Subdiscipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | NASA CENTER | **Phone:** | 281-483-2318 |
| **Organization Name:** | NASA Johnson Space Center | | |
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| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77058 | **Congressional District:** | 9 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | NOT AVAILABLE |
| **Start Date:** | 09/01/2003 | **End Date:** | 09/30/2005 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 20 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | Whitmore, Mihriban ( Habitability and Human Factors Office ) | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | Since the inception of long-duration human spaceflight, stowage management and accessibility has been reported as a major issue facing crews. Even with its relatively time in orbit, Skylab experienced stowage complications at the end of its functional life. The Mir crewmembers identified stowage as one of the primary issues that would face future space stations. Even with only three Expedition crews having provided feedback on the habitability of ISS, stowage has already been identified as an issue with every crew. The collection of feedback and crew comments collected over this history indicates that a simplified stowage system that provides optimized use of volume available, strategic organization, and easy crew access would eliminate many of the stowage problems encountered by the crew, as well as significantly increase crew productivity and safety. The project will first define and address the current on-orbit stowage situation as a method by which to scope to problem. Based on the recommendations provided by the team as a result of the assessment of the current situation, conceptual designs of a simplified stowage system will be developed. Following | | |

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| **Task Description:** | a heuristic usability assessment, prototypes of the concepts will be built for future testing. Referring back to the recommendations generated as part of this project, as well as previous spaceflight experience, the team will assess the strengths and weaknesses of the designs based on the prototypes. In addition, a formal usability test will be conducted. A final evaluation with the end user—the crewmembers—will be conducted to validate previous recommendations and gain additional feedback based on zero-gravity experience. Implementation of a stowage system that would maximize the usable dedicated stowage volume on ISS, as well as provide a more intuitive organizational scheme and better visual access to the crew, will improve crew efficiency and productivity on a daily basis. Develop human factors requirements and conceptual designs for stowage of cargo and hardware for long-duration space missions. Development of light-weight, flexible and reconfigurable stowage system that will maximize habitable volume and increase crew productivity. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Earth benefit has not been identified at this point in the progression of the specified technical development project tasks. |
| **Task Progress:** | Overall stowage analysis report has been completed for the existing space program as well as historical reference of both domestic and international space vehicles. Preliminary human factors requirements have been identified. Two case domains have been identified as candidates for concept development. |
| **Bibliography Type:** | Description: (Last Updated: 01/26/2005) |
| **Presentation** | Susan Baggerman SF3 - NASA Janis Connolly SF - NASA Lena Norris SF3-Lockheed Martin "Concepts for a Simplified ISS Stowage and Retrieval System" Habitation 2004 Conference on Space Habitation Research and Technology Development<br>Jan-2004 |